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Identification of cis-regulatory elements provides insights into tissue-specific gene regulation in the sheep genome

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Shangqian Xie[1,2], Kimberly M. Davenport[1,3], Mazdak Salavati[4], Alex Caulton[5], Emily L. Clark[4], Alan L. Archibald[4], Shannon M. Clarke[5], Rudiger Brauning[5], Alisha T. Massa[6], Michelle R. Mousel[7], Stephen N. White[6,8], Kim C. Worley[9], Gordon K. Murdoch[3], Gabrielle Becker[1], Morgan R. Stegmiller[1], Sarem F. Khilji[1], Katie A. Shira[1], Temitayo A. Olagunju[1], Tracy Hadfield[10], Stephanie D. McKay[2], Benjamin D. Rosen[11], Noelle E. Cockett[10], Timothy P. L. Smith[12] & Brenda M. Murdoch[1]

Annotation of regulatory elements is essential for understanding mechanisms underlying gene regulation, particularly tissue-specific regulation in human and animals. Here, we characterize 274,682 enhancers and 25,975 promoters across 24 tissues from an adult female sheep using ChIP-seq, ATAC-seq, CAGE-seq, RRBS, WGBS, and RNA-seq. We identify seven neural development-related genes with over 10 enhancers in brain tissues, highlighting the role of tissue-specific regulation. Cis-regulatory enhancer-promoter combinations provide insights into tissue-specific enhancers, such as the cerebellum-specific enhancer (*chr15*: *57390520-57390685*) regulating *BDNF*, which is expressed in both the cerebellum and cerebral cortex. Comparative analysis of enhancer-promoter combinations in human, mouse, pig, cattle, and sheep reveals ruminant-specific pathways, including pentose catabolism and long-chain fatty acid import regulation. A milk fat yield quantitative trait locus (QTL) identified within an enhancer interacts with the fat metabolism-related gene *COMMD1*, and a birth weight-associated QTL detected within a cerebellum-specific enhancer regulates *XKR4*. This study provides a robust framework for exploring cis-regulatory mechanisms and tissue-specific regulation, advancing the functional annotation of the sheep reference genome.

The annotation of regulatory elements in the genome and assessment of their function in various tissues are essential for understanding gene regulation and interpreting the potential impact of genomic variants on phenotype. The encyclopedia of DNA elements (ENCODE) project annotated functional sequences on a genome-wide scale in human cells and tissue types[1], and subsequently in model organisms[2,3]. These studies identified similarities and differences in regulatory elements across species. Functional annotation involves identifying and

[1]Department of Animal, Veterinary, & Food Sciences, University of Idaho, Moscow, ID, USA. [2]Division of Animal Sciences, University of Missouri, Columbia, MO, USA. [3]Department of Animal Sciences, Washington State University, Pullman, WA, USA. [4]The Roslin Institute and Royal (Dick) School of Veterinary Studies, University of Edinburgh, Midlothian, UK. [5]AgResearch, Mosgiel, New Zealand. [6]Department of Veterinary Microbiology and Pathology, Washington State University, Pullman, WA, USA. [7]USDA, ARS, Animal Disease Research Unit, Pullman, WA, USA. [8]GENUS PLC, Wisconsin, WI, USA. [9]Baylor College of Medicine-Human Genome Sequencing Center, Houston, TX, USA. [10]Animal, Dairy, & Veterinary Sciences, Utah State University, Logan, UT, USA. [11]USDA, ARS, Beltsville Agricultural Research Center (BARC), Beltsville, MD, USA. [12]USDA, ARS, U.S. Meat Animal Research Center (USMARC), Clay Center, NE, USA.
e-mail: bmurdoch@uidaho.edu

annotating the functional sequences in a genome, which include both expressed sequences (coding and non-coding RNAs) and regulatory sequences. Early research primarily focused on identifying expressed sequences using technologies like ESTs and cDNA libraries[4], laying the groundwork for understanding gene expression profiles. The advent of RNA-seq then provided a more comprehensive view of the transcriptome, quantifying gene expression across tissues and developmental stages[5,6], thereby enhancing the functional annotation of genomes. In contrast, identifying regulatory sequences on a genome-wide scale has been more limited, largely concentrated on humans and a few model organisms through projects like ENCODE[1]. These initiatives have created extensive datasets that illuminate the roles of regulatory elements, such as enhancers, promoters, and insulators, in shaping gene expression and driving complex phenotypes[7]. Both expressed and regulatory sequence annotation are paving the way for a more complete understanding of genome function.

Recently, significant progress has been made in the study of regulatory sequences in animal genomes, particularly in livestock species such as cattle[8], pig[9], and chicken[10] through the project Functional Annotation of Animal Genomes (FAANG). In cattle, research identified associations between weak enhancers, active transcription start sites (TSS), and economic traits during rumen development[8]. In pigs, multi-omics data were used to annotate chromatin states across 14 tissues, revealing the importance of active promoters and enhancers in complex traits and domestication, with distinct regulatory patterns between Asian and European pigs[9]. Similarly, researchers annotated 1.57 million regulatory elements in chicken, predicting some enhancer-gene combinations and super-enhancers, providing valuable resources for poultry genetics and genomics research[10]. These projects not only deepen our understanding of genetic regulation but also fuel comparative and evolutionary studies, as well as enabling the investigation of the genetic control of phenotypes. As an important livestock species, the study of regulatory elements in sheep is equally valuable. The ovine FAANG project offers a unique opportunity to characterize transcriptional regulatory elements in sheep, advancing our understanding of the biological mechanisms underlying complex traits in this species. Recently, we initiated the characterization of regulatory elements in three tissues (liver, spleen, and cerebellum) using chromatin immunoprecipitation sequencing (ChIP-seq) data to define chromatin states, including active and poised enhancers and promoters[7]. Other studies have identified regulatory elements in sheep macrophages, cartilage, cecum, pituitary gland, hypothalamus, liver, longissimus dorsi, rumen, spleen, and tail fat tissues[11,12]. However, current annotations remain limited to a few tissues, especially in the sheep genome. A comprehensive, systematic study identifying regulatory elements across multiple tissues is urgently needed, particularly focusing on enhancers, promoters, and their interactions.

Emerging research increasingly highlights the regulatory impact of enhancer-promoter interactions on gene expression, revealing their dynamic nature and marked tissue- and cell-type specificity[13]. A common approach for identifying enhancer regulations involves correlating chromatin signals at enhancers with gene transcriptional activity. Studies have utilized correlations between chromatin signals (H3K4me1 and H3K27ac) and gene expression to predict enhancer–promoter interactions. This correlation-based approach has proven an effective strategy for mapping regulatory interactions in research. For instance, Lin et al. identified putative enhancer-gene interactions within the same topologically associated domain based on significant positive correlations between H3K27ac signals at enhancers and gene expression[14]. Furthermore, transcription signatures, such as bidirectional and unidirectional transcription captured by cap analysis gene expression and deep sequencing (CAGE-seq), have become defining marks of active enhancers and promoters[15]. The FANTOM5 project utilized CAGE expression profiling to catalog enhancers across diverse human tissues and cell types, providing expression-based support for enhancer-promoter pairing that collectively modulate target gene expression[16]. Tissue-specific enhancer activity provides compelling examples of how these interactions underpin genomic regulation. Enhancer regions in the human prefrontal cortex selectively activate *FOXP2*[17], a gene crucial for neural development and vocal communication. In the mouse, muscle-specific enhancer regions marked by H3K27ac interact with *MYF5* during myogenesis[18], promoting muscle-specific expression. Similarly, in cattle, muscle-specific enhancers marked by H3K27ac target *Myod*, regulating muscle growth in skeletal muscle cells[19]. These reports highlight the pivotal role of enhancer-promoter or enhancer-gene interactions in shaping tissue-specific gene expression, offering valuable insights into the regulatory mechanisms underlying the complexity of genomic regulation.

To meet the need for a comprehensive analysis of regulatory elements across diverse sheep tissues and to elucidate tissue-specific characteristics of enhancer-promoter interactions, we systematically characterized enhancers and promoters across 24 tissues in this study using six high-resolution assays: chromatin immunoprecipitation sequencing (ChIP-seq), assay for transposase-accessible chromatin using sequencing (ATAC-seq), cap analysis gene expression and deep sequencing (CAGE-seq), reduced representation bisulfite sequencing (RRBS), whole-genome bisulfite sequencing (WGBS), and RNA-seq (Supplementary Fig. 1). All tissues were derived from the same individual whose genome was sequenced and assembled de novo, resulting in the current publicly available ovine reference genome ARS_UI_Ramb_v2.0[20]. These datasets facilitated the systematic mapping of enhancers, promoters, and cis-enhancer-promoter combinations throughout the sheep genome. This extensive analysis provides critical insights into tissue-specific gene regulation and the genetic architecture of complex traits in this economically significant species.

Results

Overview of collected data

Twenty-four tissue samples from the animal used to generate the ARS_UI_Ramb v2.0 reference genome was collected in this study (Fig. 1A). These samples were used to explore chromatin states and to identify and characterize key regulatory elements, such as enhancers and promoters, through a comprehensive set of 188 high-throughput sequencing datasets (Fig. 1B and Supplementary Data 1). The datasets include chromatin immunoprecipitation sequencing (ChIP-seq) for four histone marks (H3K27ac, H3K27me3, H3K4me1, and H3K4me3) along with control samples, ATAC-seq, CAGE-seq, and RNA-seq. Each contributes a distinct insight into the intricate landscape of regulatory elements for gene regulation. A total of 12.8 billion raw reads were sequenced from 188 sequencing datasets, generating 10.4 billion mapped reads, with an average mapping rate of 80.64% for all datasets (Supplementary Data 1). The average number of reads was 49,493,667; 48,223,579; 57,509.914; 63,299,552; 50,676,069; 4,321,867; 102,757,218; and 72,333,332 for H3K27ac, H3K4me3, H3K4me1, H3K27me3, Control, CAGE, ATAC and RNA-seq data, respectively (Supplementary Data 1).

Characterization of chromatin states

ChIP-seq libraries of H3K27ac, H3K4me3, H3K4me1, H3K27me3, and Control were generated from 24 tissues and sequenced (see Materials and Methods). A total of 8,802,602 peaks were identified in all four histone marks across 24 tissues and annotated on the peaks to determine their genomic features relative to the reference genome's annotated elements (Figs. 1C and 2 and Supplementary Table 1). The majority of H3K4me1 and H3K27ac peaks were mapped to intron and intergenic regions across all tissues (Fig. 1C and Supplementary Fig. 2A, B), supporting their roles as key indicators of enhancers. Approximately half of the peaks for these two histone marks were located within intron regions (50.99% ± 5.68 for H3K4me1 and 45.07% ± 6.48 for H3K27ac), while peaks near the transcription start site (TSS) within

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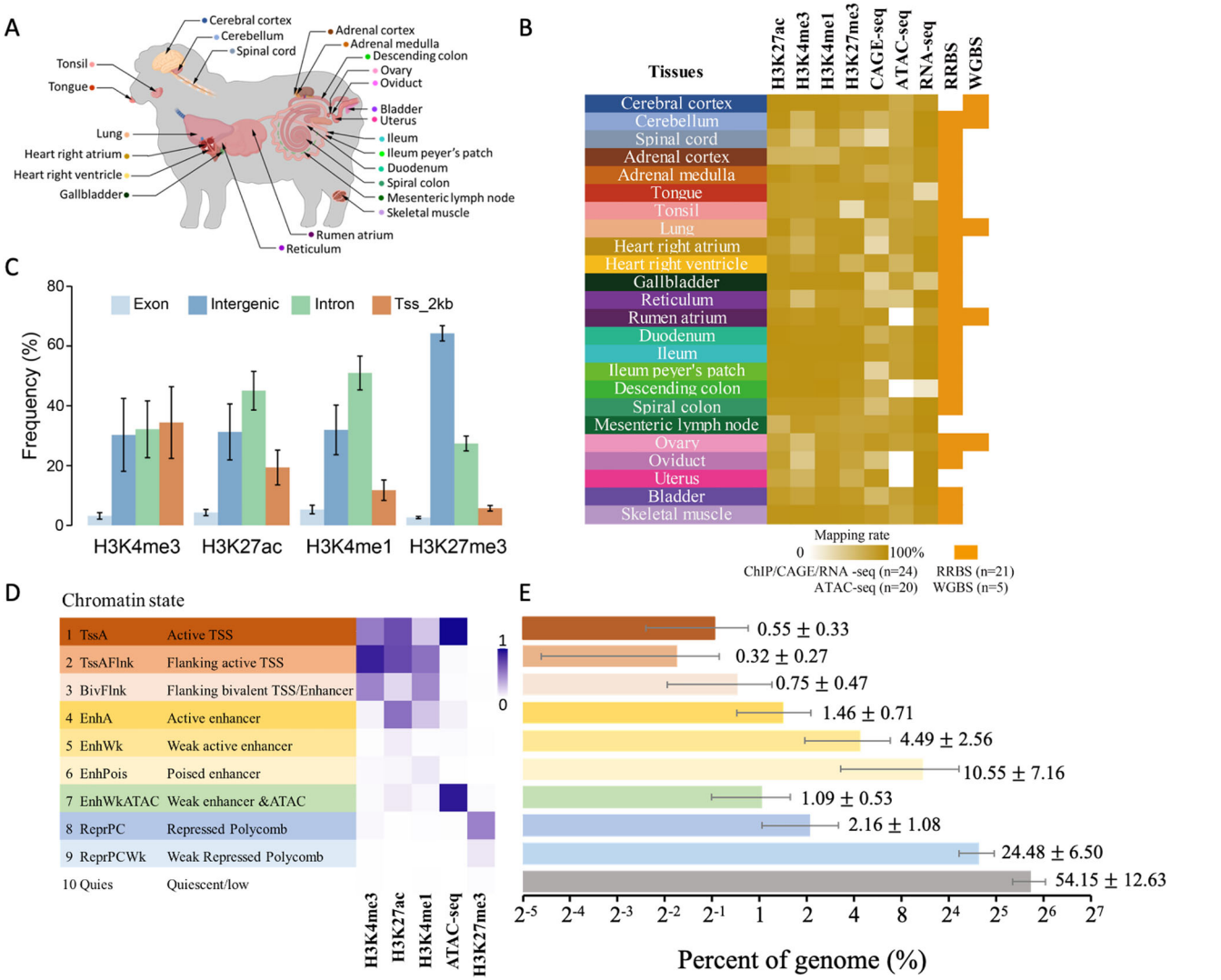


Fig. 1 | Collected sample and assay data as well as chromatin states. A Specific tissues included and **B** sequencing data assays used in this study. Schematic elements in (**A**) were created in BioRender. Xie, S. (2025) https://BioRender.com/u6bemhb. RRBS: Reduced representation bisulfite sequencing; WGBS: Whole-genome bisulfite sequencing. **C** Peak distribution of histone modification marks along the genome. Frequency was calculated for each tissue ($n = 24$). Bars represent the mean across tissues, with error bars indicating ± standard deviation (SD). Individual tissue values are provided in the Source Data file. Tss: Transcription start sites. **D** Signals of chromatin states from ChIP-seq and ATAC-seq data using the cerebellum as an example. **E** Genome occupancy of chromatin states in individual tissues ($n = 24$), shown as mean ± SD. Source data are provided as a Source Data file.

2 kb were more abundant for H3K27ac (19.37% ± 5.83) than for H3K4me1 (11.77% ± 3.41). In contrast, for H3K4me3, the majority of peaks were found near TSS regions (34.4% ± 11.99), more than in intergenic and intron (Supplementary Fig. 2C), confirming H3K4me3 and H3K27ac as characteristic marks of promoters. H3K27me3 peaks were predominantly located in regions without gene annotation (64.26% ± 2.56), with relatively few mapped near annotated TSS (5.72% ± 0.92) (Supplementary Fig. 2D). Overall, the peak localizations for all four histone marks relative to gene annotations were consistent with known properties of these marks with respect to genes and gene activity, supporting the reliability of the data for annotating promoters and enhancers and their status relative to gene activity.

Chromatin states were further annotated by ChromHMM[21], and the ten specific chromatin states were identified by using four histone marks and ATAC-seq datasets (Fig. 1D). The three states (TssA, TssAFlnk and BivFlnk) involving TSS covered ~1.6% of the genome. The active and poised enhancer states, defined by H3K27ac, H3K4me1, and ATAC signals, covered ~17.6%. Repressed polycomb and weak repressed polycomb states defined by H3K27me3 combined to cover ~26.6% (Fig. 1E). These chromatin states represented ~45.9% of the genome with distinguishable modification signals measured with these histones (Fig. 1E and Supplementary Table 2). The chromatin states from sheep were consistent with a previous study of cattle[8] and chicken[10] chromatin states.

Identification of enhancers and promoters across tissues

To further clarify the specific localization of chromatin states and identify key regulatory elements, particularly enhancers and promoters across tissues, we integrated peaks from CAGE-seq and ATAC-seq data to cross-validate those identified from ChIP-seq (Fig. 2A). By cross-validation of ChIP-seq, CAGE-seq, and ATAC-seq datasets, we identified a total of 424,559 peaks as potential enhancers and 320,425 peaks as potential promoters from 20 tissues, with four tissues excluded due to missing ATAC-seq data (Fig. 2A and Supplementary Fig. 3).

Based on the variation in the peak number of shared peaks across tissues (Supplementary Fig. 3), we further identified the high-confidence peaks identified in at least two tissues for both enhancers

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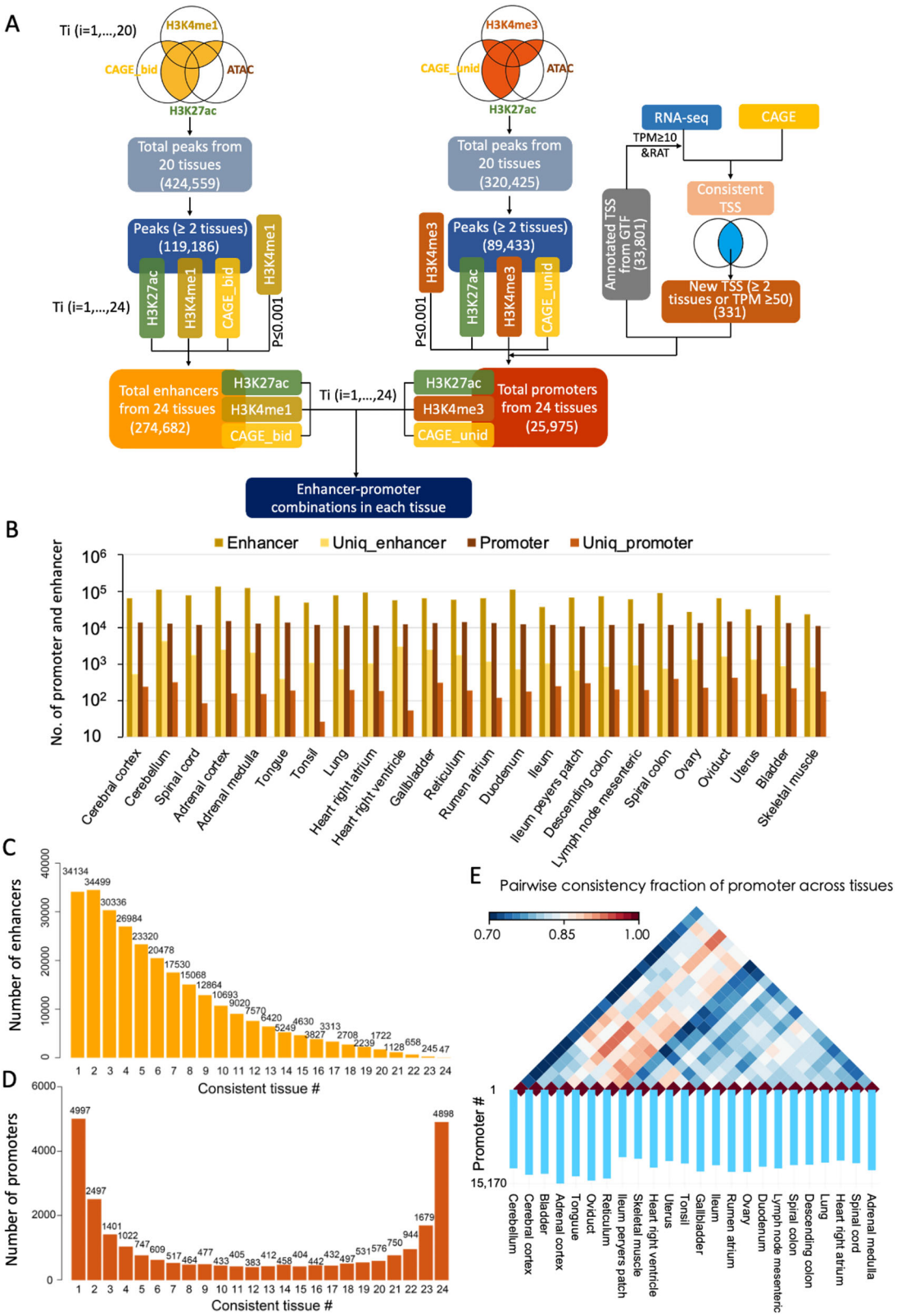


Fig. 2 | Workflow for mapping of cis-regulatory enhancers and promoters across tissues. A Analysis pipelines of the integrative strategy. TSS: Transcription start sites, GTF: Gene Transfer Format, TPM: Transcripts per million. **B** Number of enhancers, promoters, tissue-specific enhancers, and tissue-specific promoters across 24 tissues (Uniq_enhancer and Uniq_ promoter represent those uniquely identified in a single tissue), and individual tissue values are provided in the Source Data file. **C** Number of shared enhancers across 24 tissues. **D** Number of shared promoters across 24 tissues. **E** Pairwise consistency fraction of promoters across 24 tissues.

and promoters. This resulted in a total of 119,186 and 89,433 peaks, respectively (Fig. 2A and Supplementary Fig. 3), which were subsequently used to infer potential enhancers and promoters specific to each of the 24 tissues. Intersecting the peaks with H3K4me1, H3K27ac, and CAGE data yielded an average of 71,340 potential enhancers per tissue, out of a total of 274,682 regions across all tissues combined (Fig. 2B, Supplementary Fig. 4A, and Supplementary Data 2 and 3). The width of these potential enhancers ranged from 50 bp to 1000 bp (Supplementary Fig. 4B). For identification of promoter, TSSs identified from RNA-seq and CAGE-seq data, capturing both annotated and unannotated TSSs, were used to facilitate the mapping of promoters (Fig. 2A). A total of 33,801 annotated TSS were derived from the GTF file, and additional 331 unannotated TSS were identified through integrated analysis of RNA-seq and CAGE data (Fig. 2A and Supplementary Fig. 5A). After intersecting the 89,433 peaks with the TSS regions, a total of 25,975 promoters were identified across tissues, comprising 271 unannotated promoters and 25,704 annotated promoters (Fig. 2B, Supplementary Fig. 5B, and Data 2 and 4). The majority of promoters exhibited widths within a 0–500 bp range, with distances to the TSS similarly falling within 0–500 bp (Supplementary Fig. 5C, D).

In the tissues examined, the adrenal cortex exhibited the greatest number of enhancers (135,371) and promoters (15,170), with enhancer variation across tissues notably exceeding that of promoters, while the cerebellum contained a relatively large number of tissue-specific enhancers (4,320) and promoters (323), particularly tissue-specific enhancers, which were markedly enriched compared with other tissues (Fig. 2B and Supplementary Table 3). Furthermore, the distribution patterns of shared enhancers and promoters across tissues differed markedly (Fig. 2C-E and Supplementary Fig. 4C). The shared enhancers numbers declined as the number of tissues increase (Fig. 2C), while promoters show a bimodal distribution, with both tissue-specific and broadly shared promoters more prevalent (Fig. 2D). Across all tissue pairs, the pairwise consistency fraction of promoters exceeded 0.7, substantially higher than that observed for enhancers (0.1) (Fig. 2E and Supplementary Fig. 4C). These findings suggest that promoters are less tissue-specific than enhancers, indicating that different tissues may have unique enhancers, which was consistent with the observation that enhancer variability across tissues is greater than that of promoters, and also supports the relatively high average of enhancers per promoter (Fig. 2B and Supplementary Table 3). Additionally, 271 unannotated promoters were identified across 24 tissues, exhibiting a similar bimodal distribution (Supplementary Fig. 5E and Supplementary Data 4). Interestingly, among these, nine tissue-specific TSSs were aligned with annotated gene sequences, of which five were annotated in other species (*Bos Taurus*, *Capra hircus*, and *Cervus elaphus*), implying that they may correspond to unannotated genes in sheep (Supplementary Data 5).

Cis-enhancer and promoter combinations across tissues

To further explore the relationship between gene enhancers and promoters, we identified enhancer-promoter combinations based on correlations of gene expression and enhancer activity. Out of 25,975 genes with identified promoters, 21,507 of them were expressed, and 4468 were not expressed in any tested tissue. We identified a total of 10,197 combined cis-enhancer-promoter regulatory elements involving 4293 genes and 8860 enhancers across 24 tissues (Supplementary Data 6). Among these, 2607 were supported by the identified cis-enhancer and promoter combinations from CTCF or CAGE data (see M&M), involving 1180 genes (Supplementary Data 6). For expressed genes, gene expression abundance (Transcripts per million, TPM) differed significantly across tissues with enhancers compared to those without, with higher expression of genes observed in tissues with enhancers than in those without (Fig. 3A). Furthermore, in tissues where a cis-enhancer was present (colored in Fig. 3B), the corresponding genes were predominantly expressed, whereas non-expressed genes consistently lacked an associated enhancer in these tissues (Fig. 3B and Supplementary Fig. 6). For example, the gene *ADORA1* on chromosome 12, which encodes the adenosine A1 receptor with important functions in the brain. Two enhancers *Enh1* (*Chr12: 682,652-683,139*) and *Enh2* (*Chr12: 737,891-738,081*) were paired with *ADORA1*, both showing a high correlation with gene expression across tissues. This gene was expressed in three brain-related tissues (cerebellum, cerebral cortex, and spinal cord). *Enh2* was consistently identified in these tissues (correlation = 0.998, $R^2 = 0.996$). Interestingly, *Enh1* associated with *ADORA1* was also detected in the bladder (correlation = 0.844, $R^2 = 0.698$) (Fig. 3C). This aligns with previous research highlighting the role of luminal adenosine turnover and A1 receptor activation in regulating bladder function[22], providing additional validation for our analysis.

Promoter and enhancer hubs reveal two predominant interaction patterns in gene transcription regulation, where enhancer hubs involve multiple promoters connected to a single enhancer, and promoter hubs involve multiple enhancers connected to a single promoter. In both hubs, the most common interaction type is a single enhancer paired with a single promoter (Supplementary Fig. 7). Approximately 52% of promoters were associated with a single enhancer (Supplementary Fig. 7A), while 89% of enhancers associated with a unique promoter (Supplementary Fig. 7B). These interaction patterns observed across all tissues were consistent with the trends in each tissue-specific hub (Fig. 3D, E and Supplementary Data 7). Notably, the vast majority of genes (99%) in promoter hubs interacted with fewer than ten enhancers (Fig. 3D), with most enhancers regulating only one gene, a smaller subset regulating two to three genes, and very few regulating more than three (Fig. 3E). Interestingly, brain and adrenal-related tissues exhibited similar hub patterns within each cluster (Fig. 3F), with more than 10 enhancers consistently identified for seven genes (*DNAJC6*, *FMNL2*, *NAV3*, *AMER2*, *NCAM1*, *OTUD7A*, and *LRRTM3*) in both brain-related tissues (cerebellum and cerebral cortex). These genes play critical roles in neural development and function: *DNAJC6* encodes an *HSP40* chaperone protein that regulates ATPase activity and metabolism; *FMNL2* governs cell migration, size, and shape changes; *NAV3* is essential for normal nerve development, with both overexpression and under-expression linked to neural deficits; *AMER2* functions as a tumor suppressor via the WNT pathway; *NCAM1* is involved in cell adhesion, development, and synaptic regulation; *OTUD7A* is crucial for neural development; and *LRRTM3* likely supports normal neuronal function and has been implicated in Alzheimer's disease. The presence of multiple enhancers regulating these genes highlights their critical importance in maintaining proper neural development and function. This intricate modulation by enhancers ensures precise gene expression during critical windows of development while minimizing the risk of dysregulation-associated pathologies. The observed greater diversity and variability in enhancers compared to promoters aligns with known biological processes, reflecting the dynamic and tissue-specific roles of enhancers in gene regulation.

Tissue-specific enhancer-promoter combinations and functional annotation

To investigate the functional roles of enhancers and promoters, we identified and compared the genes linked to tissue-specific promoters and enhancers from enhancer-promoter combinations. Notably, the genes associated with tissue-specific promoters were all included in the genes of enhancers from tissue-specific enhancer-promoter combinations (Fig. 4A and Supplementary Data 8). The tissue-specific enhancer-promoter combinations provide additional insights, as they incorporate interactions of enhancers and promoters that are not captured when considering tissue-specific gene expression alone. While previous analyses have focused on promoter-gene specificity derived from gene annotations, the inclusion of the unrecognized

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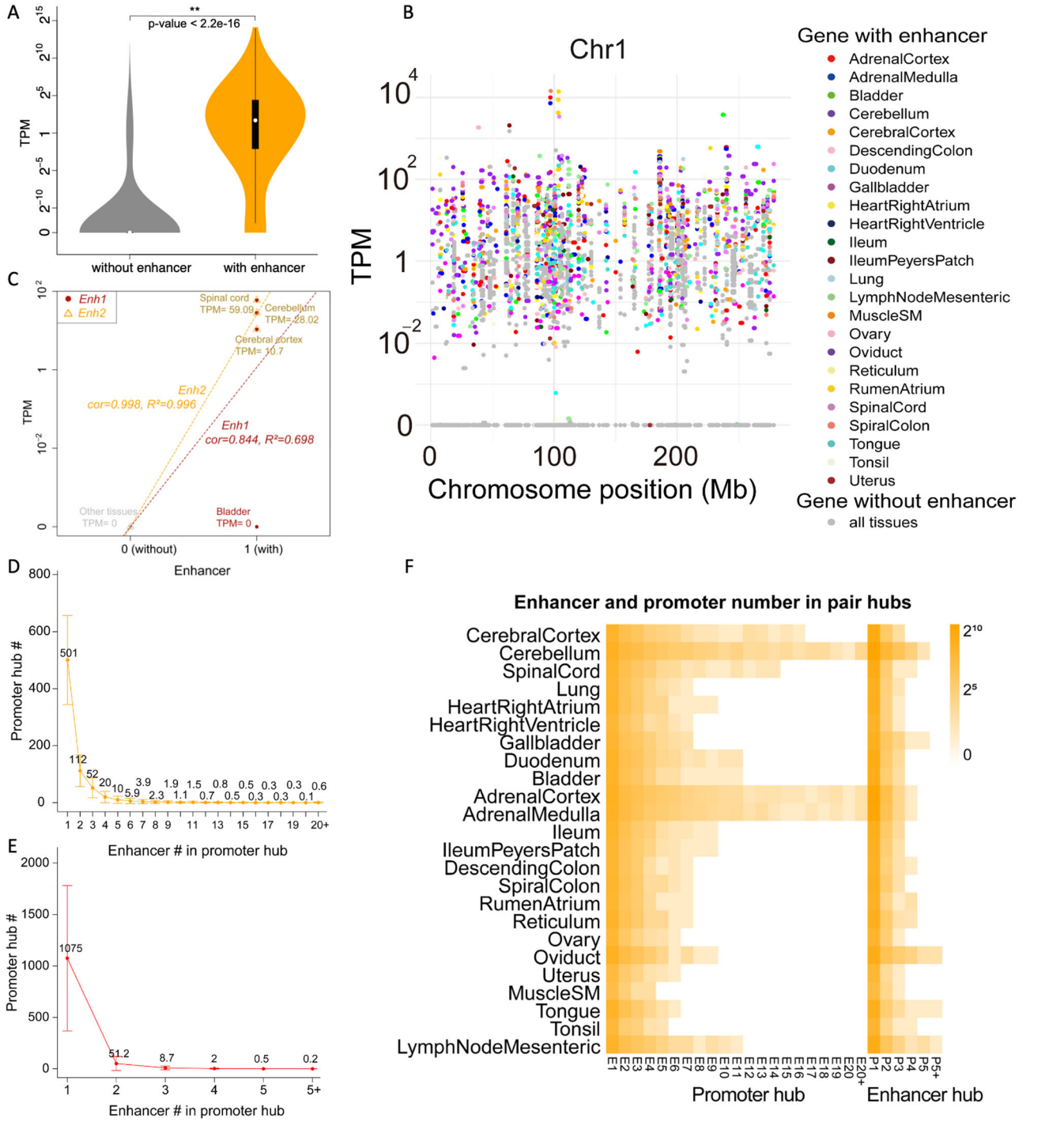


Fig. 3 | Identification of cis-enhancer-promoter combination across tissues. **A** Comparison of gene expression between genes with and without an enhancer across tissues ($n = 24$). Box plots show the mean (white point), the 25th and 75th percentiles (bounds of the box), and whiskers extending to the minimum and maximum values. Statistical significance was assessed using a two-sided Wilcoxon rank-sum test: $P < 2.2 \times 10^{-16}$. Gene expression values for each tissue are provided in the Source Data file. **B** Relationship between gene expression and enhancer in each tissue along chromosome 1 (Chr1). TPM: Transcripts per million. **C** Correlation between two enhancers of gene *ADORA1* and their expression across tissues. Enh1 enhancer1, Enh2 enhancer2. **D** Mean and standard deviation of the number of enhancers in promoter hubs across tissues ($n = 24$). **E** Mean and standard deviation of the number of promoters in enhancer hubs across tissues ($n = 24$). Tissue-level hub values underlaying panels **D**, **E** are provided in the Source Data file. **F** Distribution of enhancer and promoter numbers in pair hubs across tissues (The X-axis shows the number of enhancers (1, 2, 3, …, 20, and >20) per promoter hub and the number of promoters (1, 2, 3, 4, 5, and >5) per enhancer hub. The heatmap represents the counts of promoter hubs and enhancer hubs in each tissue, with the indicated number of enhancer or promoter connections.

regulatory roles of enhancers from enhancer-promoter combinations uncovers insights into gene regulation.

To validate the contribution of enhancers from combinations, we performed GO enrichment analysis on genes associated with tissue-specific enhancer genes and compared with these to promoter-associated genes. GO terms reflected the biological functions of the respective tissues, with particularly strong associations observed in brain and heart tissues (Fig. 4B and Supplementary Data 9). Among the

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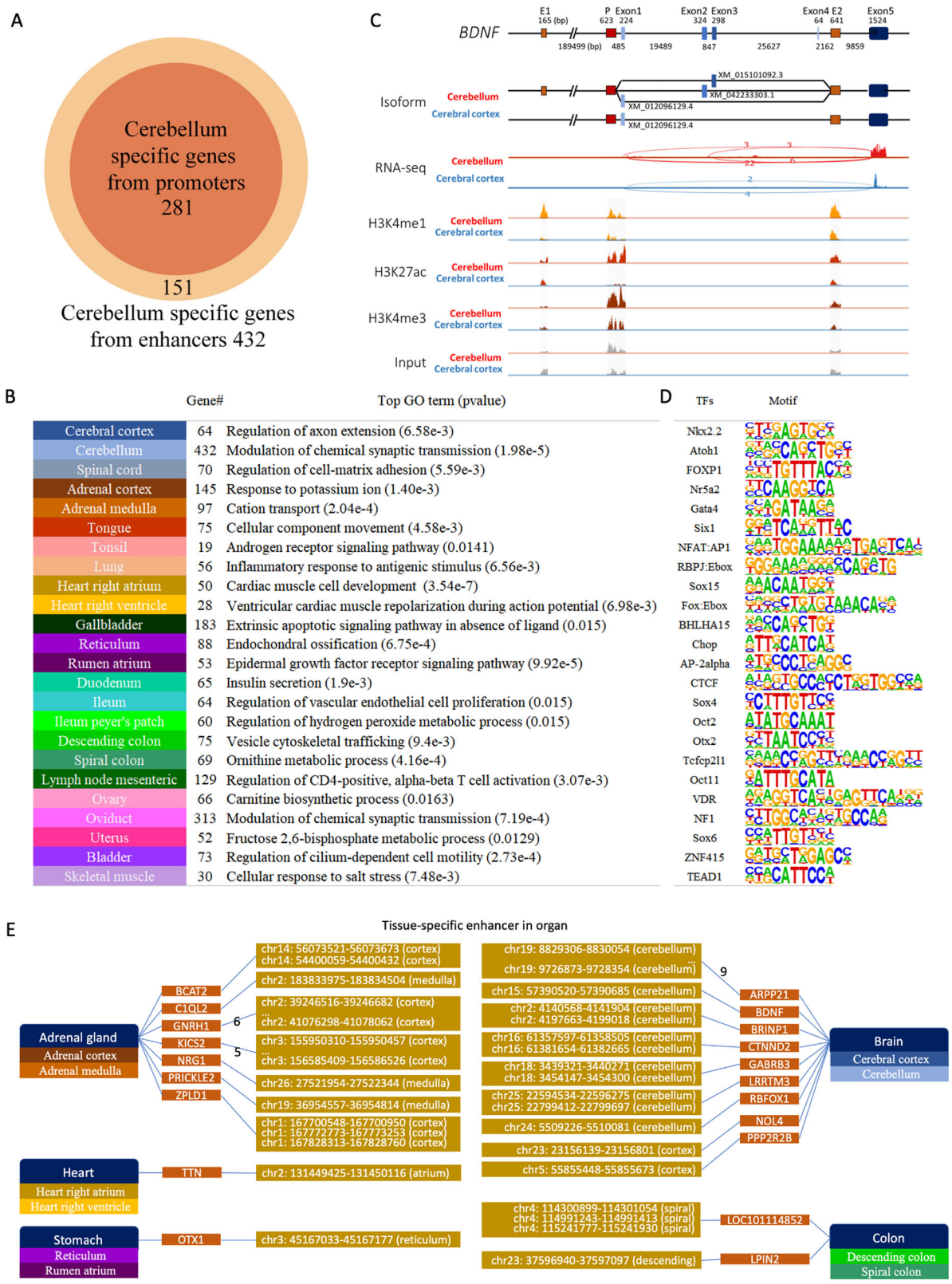


Fig. 4 | Tissue-specific enhancers, GO enrichment, and motifs of these specific sequences. A Associated genes from specific promoters and enhancers in the cerebellum, as an example (other tissues are included in Supplementary Data 8). **B** GO enrichment of tissue-specific enhancer sequences. **C** Annotation and validation of enhancer and promoter in the *BDNF* gene using RNA-seq and ChIP-seq data as evidence to support tissue specificity. XM_, E, and P in the isoform track are isoforms, enhancer, and promoter, respectively. The values in the RNA-seq track indicate the counts of junction reads spanning different exons. H3K4me1, H3K27ac, H3K4me3, and Input tracks are the visualization of peaks; exon 4 is absent from the three brain transcripts but present in other annotated transcripts. **D** TF and Motif of tissue-specific enhancer sequences. **E** Representative tissue-specific enhancer for the selected genes within the same organ type (numbers 5, 6, and 9 indicate the count of specific enhancers of that gene, with the full list provided in Supplementary Data 13).

24 tissues analyzed, genes associated with both enhancers and promoters displayed overlapping GO terms in 14 tissues. However, in the remaining ten tissues, GO analysis using promoter-enhancer combinations yielded stronger associations, including lower p values, a greater number of GO terms, and higher tissue correlation, compared to results obtained using promoter-only information (Supplementary Data 10). Furthermore, we performed GO analysis specifically for the 151 genes uniquely associated with enhancers in the cerebellum (Fig. 4A), five significant biological processes were identified and related to synaptic transmission and neuronal differentiation (Supplementary Table 4). Notably, 11 key genes (e.g., *BDNF*, *NLGN3*, and *NRCAM*) were shared across multiple pathways (Supplementary Fig. 8), highlighting their central roles in neuronal development and synaptic function, and reinforcing the biological relevance of the enhancer-specific gene set. Moreover, the cerebellum-specific enhancer-promoter combination involving the *BDNF* gene further exemplifies this regulatory relationship (Fig. 4C and Supplementary Data 11). Although *BDNF* is expressed in both the cerebellum and cerebral cortex, its cerebellum-specific expression was not detectable using promoter-gene specificity alone. By incorporating enhancer data, we identified a cerebellum-specific enhancer (Enh1, *chr15: 57,390,520-57,390,685*) active exclusively in the cerebellum, whereas a second enhancer (Enh2, *chr15: 57,150,237-57,150,878*) was shared between the cerebellum and cerebral cortex. This discovery yields a reasonable hypothesis that the cerebellum-specific enhancer may be the key to tissue-specific *BDNF* expression relevant to the distinct cerebellar phenotypes when compared to the cerebrum. Differential H3K4me1 signals in Integrative Genomics Viewer (IGV) from ChIP-seq data further validated tissue-specific enhancer activity (Fig. 4C). RNA expression of exon and read junctions from RNA-seq revealed three *BDNF* isoforms in the cerebellum, of which two were exclusive to this tissue (XM_015101092.3 and XM_042233303.1), compared to one isoform in the cortex. These findings suggest a regulatory association between enhancer Enh1 (*chr15: 57,390,520-57,390,685*) and cerebellum-specific isoforms of *BDNF* (Fig. 4C).

To further elucidate enhancer function, we analyzed tissue-specific enhancer sequences to identify associated transcription factors (TFs). A total of 81 TFs were identified (Fig. 4D and Supplementary Data 12), many of which are known regulators of the gene expression of corresponding tissues. For instance, *Nkx2.2*, identified in the cerebral cortex, is critical for oligodendrocyte differentiation and CNS development, and *Atoh1* from cerebellum tissue plays a pivotal role in neurogenesis. Forkhead box (*Fox*), prominently associated with cardiac tissues, cardiovascular development, cell differentiation, and stress response. Similarly, *SOX15* from SOX family TFs contributes to the development and structural integrity of cardiac tissue (Fig. 4D). These results highlight the importance of enhancers in tissue-specific regulation and emphasize their role in defining enhancer-promoter interactions. These discoveries establish opportunities for continued hypothesis-based research that will elucidate the nuances of tissue-specific phenotypes established through tissue-specific gene expression regulation.

To account for organ-level specificity, we extended our analysis to identify enhancers associated with broadly tissue-specific combinations within key organs, including the brain, heart, adrenal gland, stomach, and colon. By analyzing tissues derived from the same organ, we aimed to explore how different tissues within an organ utilize distinct enhancers to regulate the expression of shared genes, thereby highlighting tissue-specific regulatory mechanisms within an organ. This expanded analysis revealed that although tissues from the same organ often share target genes regulated by enhancers, different tissues frequently employ distinct enhancers for these genes, reflecting tissue-specific regulatory diversity within the organ (Fig. 4E and Supplementary Data 13). For instance, in the brain, enhancers specific to the cerebellum and cerebral cortex were found to regulate common neural genes, such as *BDNF*, but the specific enhancers used differed between these two tissues, as previously discussed. Similarly, in both the stomach and colon, distinct enhancers targeting shared genes involved in epithelial barrier maintenance and digestive enzyme production emphasized tissue-level specialization within the gastrointestinal system. These findings demonstrate that, even within the same organ, the regulatory landscape is finely tuned to the unique functional requirements of individual tissues and likely at very specific timing and life stages. The identification of such enhancer combinations not only underscores the complexity of gene regulation at the organ level but also provides valuable insights into the mechanisms driving tissue specificity within broader organ systems. This enhanced understanding of intra-organ regulatory dynamics offers a critical foundation for future studies on tissue- and organ-specific gene regulation.

Tissue-specific enhancer-promoter combination and methylation

To explore the specific impact of enhancer-promoter combinations on gene expression across tissues, we collected DNA methylation data from whole-genome bisulfite sequencing (WGBS) and reduced representation bisulfite sequencing (RRBS) across 22 tissues (Fig. 1B) and analyzed the association between DNA methylation within enhancer-promoter combinations and the RNA expression of related genes. Results from both WGBS (all five tissues) and RRBS (21 tissues) revealed that methylation levels in enhancer regions were negatively correlated with the expression of their cis-regulatory genes (Fig. 5A and Supplementary Figs. 9A and 10), highlighting the distinct regulatory roles of enhancers in gene expression. For promoter regions, RRBS methylation levels did not exhibit a clear correlation with gene expression (Supplementary Fig. 11). However, methylation patterns derived from WGBS (five tissues) were consistently negative (Fig. 5B and Supplementary Fig. 9B), aligning with previous studies reporting that promoter methylation decreases gene expression[23]. The findings indicated that the repressive effect of enhancer methylation on gene expression is consistent with that of promoter methylation. This study demonstrates the suppressive impact of enhancer methylation on gene expression in sheep, which aligns with known biological mechanisms. Methylation likely significantly influences the interaction between enhancers and promoters, thus regulating gene activation. This commonality in methylation effects on both enhancers and promoters is biologically plausible and aligns with findings from previous studies[24], highlighting the role of methylation in impairing enhancer-promoter interactions. Furthermore, RRBS data revealed that enhancer methylation exhibits a stronger repressive effect on gene expression than promoter methylation, consistent with findings in human cancer studies[25].

Cross-validation of cis-enhancer-promoter combinations across species

To further substantiate the functionality of the identified enhancers and promoters in sheep, we conducted a comparative analysis of four selected tissues (cerebellum, cortex, lung, and muscle) across five species (Fig. 6A). Relative to human, we identified an average of 8914 promoters in sheep tissues, compared to 7736 in cattle, 7963 in pig, 5313 in mouse, and 21,792 in human. Enhancers were identified more than promoters, averaging 42,904 in sheep, 17,808 in cattle, 18,770 in pigs, 123,066 in mouse, and 280,892 in humans (Supplementary Data 14).

Interspecies comparison of enhancers revealed that 61%, 68%, 68%, and 72% of sheep enhancers in cerebellum (69,399), cortex (42,467), lung (46,371), and muscle (13,378), respectively, were conserved in at least one other species. (Fig. 6B and Supplementary Fig. 12A-C). Promoters exhibited even higher conservation rates across species, with 86, 89, 88, and 92% of promoters in the same tissues

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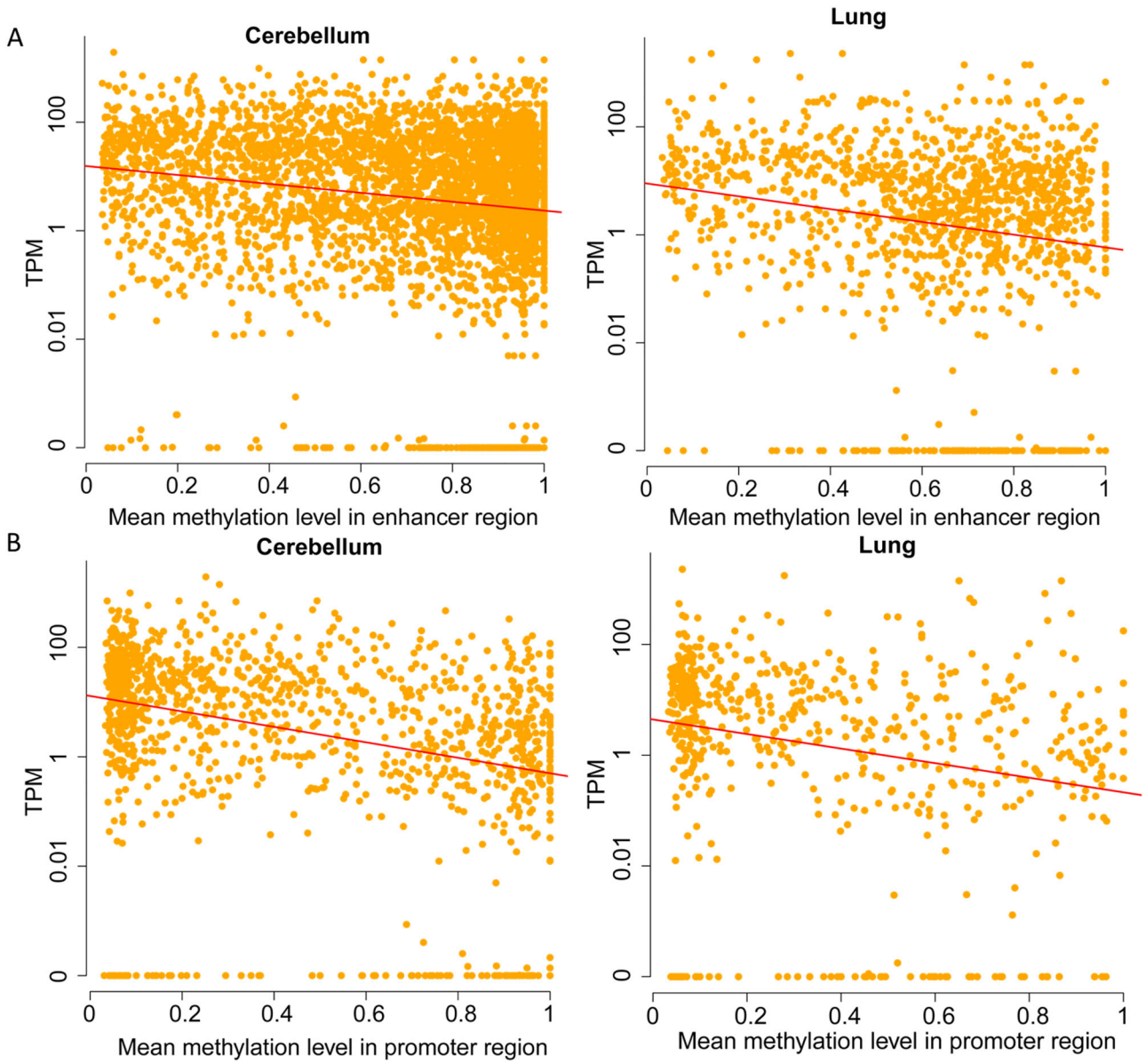


Fig. 5 | Methylation in enhancers and promoters correlates with gene expression. A Enhancers in cerebellum and lung (others in Supplementary Fig. 9A). **B** Promoters in cerebellum and lung (others in Supplementary Fig. 9B).

conserved in other species (Fig. 6B and Supplementary Fig. 12D-F). Moreover, on average, approximately 17% of promoters (17.4% in cerebellum, 17.7% in cortex, 16.5% in lung, and 18.3% in muscle) were conserved across all five species, substantially higher than enhancers 0.7% (0.6, 0.6, 0.7, and 1% in the respective tissues) (Fig. 6B and Supplementary Fig. 12). Additionally, sheep exhibited more conservation with cattle than pig, reflecting the closer evolutionary relationship between these ruminant species. (Fig. 6B and Supplementary Fig. 12). Notably, enhancers showed higher overlap observed for human and mouse may result from their more comprehensive annotation and the larger catalog of regulatory elements available for these species. Promoters displayed consistently higher conservation rates across all five species compared to enhancers, highlighting a fundamental difference in conservation patterns between the two types of regulatory elements (Fig. 6C, Supplementary Fig. 13, and Supplementary Data 15), which is consistent with previous observations in mammalian species[26]. These cross-species comparisons underscore the greater variability of enhancers relative to promoters, emphasizing the enhancer regulatory specificity and functional diversity in sheep.

We further focused on tissue-specific cis-enhancer-promoter combinations, we identified 596, 107, 117, and 23 unique combinations in cerebellum, cortex, lung, and muscle, respectively (Fig. 6D). Functional enrichment analysis of these cis-regulatory elements revealed strong associations with tissue-specific biological functions: cerebellum-related combinations were significantly enriched in synaptic organization, cortex combinations in MAPK cascade regulation, lung combinations in smooth muscle apoptosis regulation, and muscle combinations in G protein-coupled opioid receptor signaling pathways (Fig. 6E and Supplementary Data 16). Furthermore, the cerebellum and cortex tissues exhibited the highest gene overlap (178 genes), with functional terms pointing to brain-specific roles, including nervous system development and synaptic transmission (Fig. 6E and Supplementary Data 16). Additionally, 69 gene-associated enhancer-promoter combinations conserved exclusively in ruminants (sheep and cattle) (Supplementary Data 17), or for which the GO enrichment was relevant to ruminant physiology, such as pentose catabolic processes and regulation of long-chain fatty acid import (Fig. 6E and Supplementary Data 16). These findings validate the robustness of the

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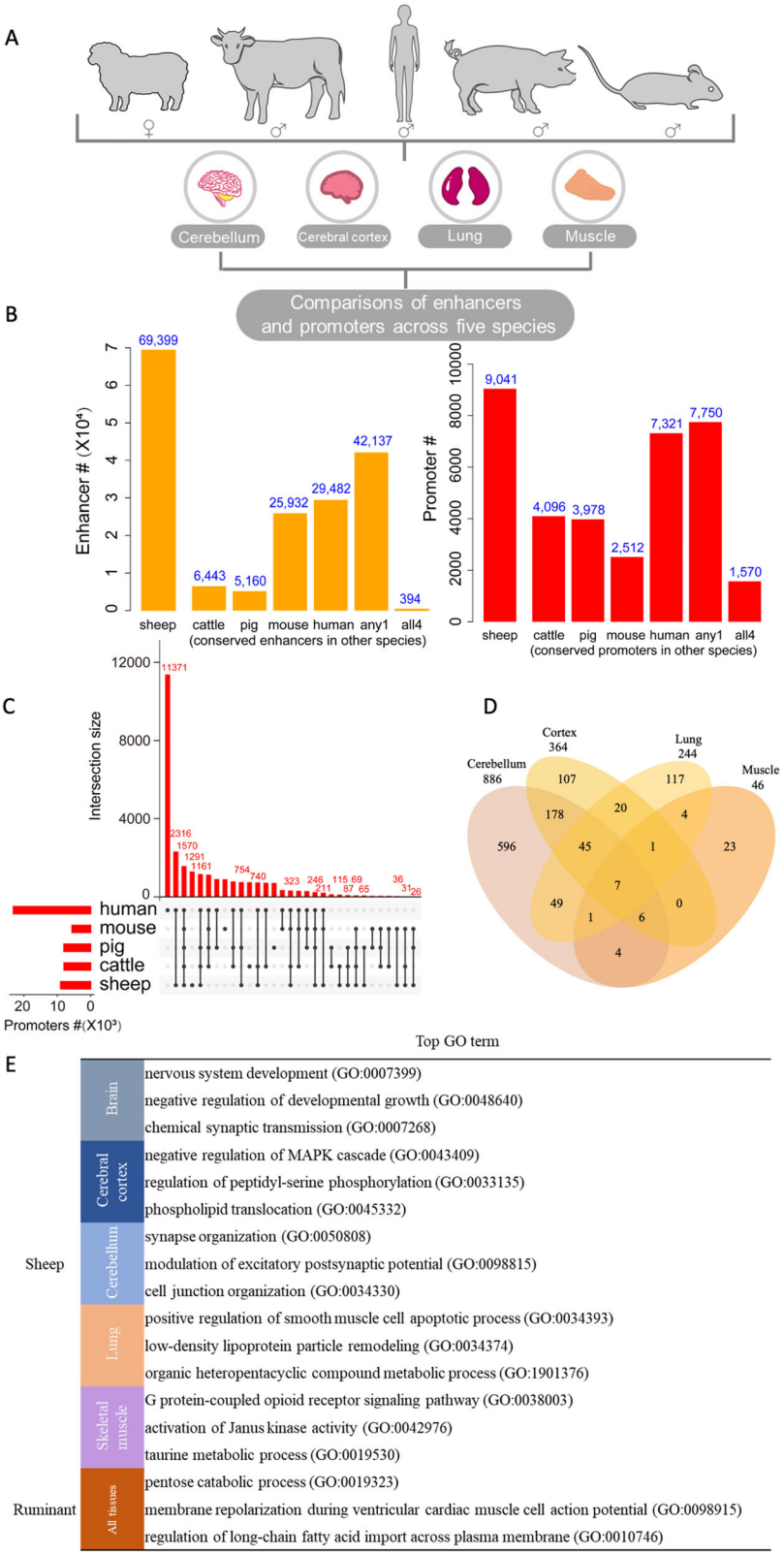


Fig. 6 | Cross-species comparison of sheep enhancers and promoters. A Tissues examined across five species, and the elements were created in BioRender. Xie, S. (2025) https://BioRender.com/u6bemhb. **B** Conservation of sheep enhancers and promoters in the cerebellum across the four other species. **C** Intersections of sheep promoters with those of other species. **D** Genes from enhancer-promoter combinations uniquely identified in sheep across four tissues. **E** Gene Ontology (GO) enrichment analysis of tissue-specific genes.

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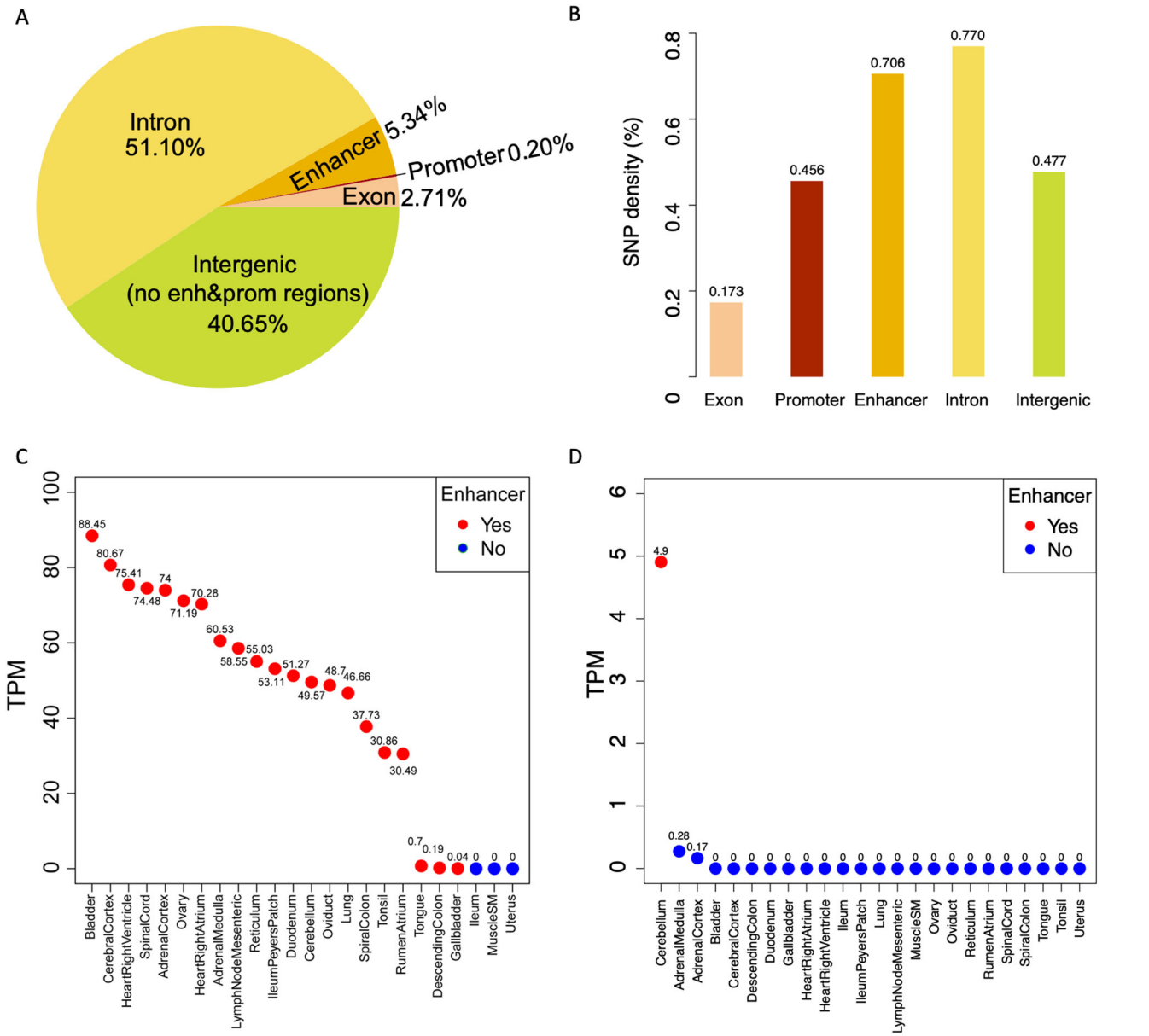


Fig. 7 | Biological insights of cis-regulatory elements with SNPs. A Genomic distribution of Single Nucleotide Polymorphism (SNP) in the sheep genome. **B** SNP density of genomic feature regions (exon, enhancer, promoter, intron, and intergenic). **C** The relationship between gene expression and the enhancer in *COMMD1*, and **D** the *XKR4* gene.

identified enhancers and cis-enhancer-promoter combinations, which effectively reflect tissue-specific regulatory functions and species-specific biological processes.

Biological insights of cis-regulatory elements with genetic variation in sheep

A total of 15,816,440 SNPs were identified from whole-genome sequencing data of 355 sheep, which were downloaded from the iSheep database (https://ngdc.cncb.ac.cn/isheep/download), of which 8% were located within exon, promoter, and enhancer regions (Fig. 7A). The number of SNPs in intergenic (without enhancer and promoter regions) and introns were significantly higher than in regulatory elements (promoters and enhancers) and exon regions, and SNP density was higher in enhancers and promoters compared to exon regions (Fig. 7B). To further explore the biological functions of SNPs within enhancer and promoter regions, we retrieved sheep QTL from the sheepQTLdb (https://www.animalgenome.org/QTLdb). A total of 38 QTLs colocalized with 17 genes in promoter regions and 215 QTLs overlapping 214 enhancers were identified (Supplementary Data 18). Among these, one QTL for milk fat yield colocalized with an enhancer interacting with the *COMMD1* gene. *COMMD1* is expressed in at least 21 tissues, with the same enhancer consistently identified across all of them (Fig. 7C). The correlation between this enhancer and *COMMD1* gene expression was strong ($r = 0.90$, $P = 1.17 \times 10^{-8}$), suggesting a potential regulatory relationship. *COMMD1* is a compelling candidate gene given its established role in copper metabolism, which has been linked to lipid metabolism and milk fat composition in ruminants[27,28].

To further assess the reliability and functional relevance of the identified cis-regulatory elements associated genes, we analyzed 23,056,026 SNPs from a recent genome-wide association study (GWAS) of birth weight (BW) in 3007 sheep[29]. Among these, 28,736 SNPs were significantly associated with BW ($P \leq 0.001$). Intersecting these variants with our enhancer–promoter interactions across 24 tissues identified 18 SNPs located within 17 enhancers active in the

cerebellum and/or cerebral cortex (Supplementary Data 19). Notably, seven cerebellum-specific enhancers were predicted to regulate genes including *XKR4*, *GALNT9*, *RET*, *TRNAC-ACA-97*, *TRNAG-CCC-197*, *TRNAC-GCA-185*, *LOC114110132*, and *LOC121817986*. Of particular interest, the most significant SNP ($P = 2.23 \times 10^{-7}$), located within a cerebellum-specific enhancer, is predicted to regulate *XKR4*. In the same GWAS study[29], *XKR4* had previously been associated with BW based on an intragenic SNP. Our analysis demonstrates a potential regulatory mechanism by which this, and an additional variant within a regulatory enhancer controlling *XKR4*, may exert regulatory control on *XKR4*, thereby providing additional evidence that variation within regulatory elements can influence phenotypic traits in sheep. In addition, *GALNT9* encodes a brain-specific glycosyltransferase with high expression in the human cerebellum[30], and *RET* is known to regulate cell growth, differentiation, and nervous system development[31]. Further, we identified another significant SNP ($P = 7.47 \times 10^{-5}$) within an enhancer active in the cerebellum and spinal cord that interacts with *TSPAN5* (Supplementary Data 19), a gene highly expressed in the brain during postnatal development in mice[32].

To obtain more direct evidence of enhancer-associated gene expression regulation, we analyzed 181 published significant eQTLs from the same study, we identified one variant significantly associated with LOC121819880 expression ($P = 4.69 \times 10^{-8}$), which resides within its interacting enhancer (*chr6:35907531-35908224*) (Supplementary Data 20). This observation provides additional evidence that sequence variants within enhancers can directly influence gene expression, reinforcing the accuracy and biological validity of our enhancer annotations. Collectively, these findings demonstrate how incorporating enhancer information enhances the biological interpretation of gene regulation in sheep and provides insights into ruminant-specific functional mechanisms. More functional validation is needed to confirm the causal roles of these enhancer-associated variants.

Discussion

Over the past decade, genomic information has revolutionized the animal breeding sector[33]. Recently, genomic research has advanced dramatically through the construction of hundreds, or even thousands, of pan-genomes and achieving telomere-to-telomere (T2T) assemblies[34,35]. Despite these significant strides in genome assembly, the functional annotation of assembled genomes has not kept pace, leaving a critical gap in our applied understanding of genome functionality. Regulatory elements, such as enhancers and promoters, play pivotal roles in orchestrating complex biological processes, and their systematic identification and characterization are fundamental to uncovering the intricate mechanisms underlying gene regulation and phenotypic diversity. Bridging this gap is essential for maximizing the utility of high-quality genome assemblies and advancing insights into functional genomics. The ENCODE and FAANG projects are two global initiatives aimed at addressing the critical gap in functional genome annotation for humans and animals[1,36], respectively. FAANG, in particular, seeks to expand the functional annotation of farmed animal genomes, covering a wide range of species, including cattle[8], sheep[7], pigs[9], and chickens[10]. As contributors to the NCBI submission of three successive updates to the sheep reference genome and as leading participants in the Ovine FAANG project[7,20], we present herein the comprehensive endeavor to systematically identify and characterize regulatory sequences within the sheep genome using an integration strategy using multi-omics data assays from 24 tissues in this study. This work is anticipated to provide substantial advancements for the sheep research community, enhancing our understanding of functional genomics and facilitating discoveries in sheep biology.

The precise identification of enhancers and promoters is fundamental to understanding the spatial and temporal specificity of gene transcription and regulation across tissues. Recent advancements in histone modification, chromatin accessibility, and chromatin structure analysis have enabled comprehensive, genome-wide identification of enhancers and promoters[10,26,37]. Specific histone modifications, such as H3K4me1 and H3K4me3, are highly indicative of enhancer and promoter regions, respectively[38]. Traditional strategies for identifying enhancers and promoters often rely on a cell/tissue-peak approach, where ChIP-seq signals (e.g., H3K4me3, H3K4me1, H3K27ac, and H3K27me3) are used to infer chromatin states in individual cell lines or tissues, and the active, poised, and repressed states are aggregated to predict potential enhancer or promoter regions within specific cells or tissues. While effective, this approach faces challenges and does not entirely consider potential tissue differences. Regulatory elements inferred from ChIP-seq peaks alone can both overestimate enhancer or promoter regions (false positives) and/or fail to identify peaks due to signal complexity resulting definitively correspond to functional elements[39,40]. Moreover, variations in peak boundaries across tissues lead to positional inconsistencies, complicating the alignment of enhancers and promoters on the reference genome[39]. These limitations are evident in existing FAANG studies on regulatory elements in cattle[8], pigs[9], and chickens[10]. Additionally, data from cap analysis gene expression and deep sequencing (CAGE-seq), and CCCTC-binding factor (CTCF) binding allow for delineation of these regulatory elements as well[10,41]. In this study, we propose an integration strategy to accurately identify genomic promoters and enhancers by integrating multi-omics assays, including ChIP-seq, CAGE-seq, ATAC-seq, and RNA-seq. This approach transcends the traditional tissue-peak signal-based identification of potential enhancers and promoters from active, poised, and repressed chromatin states, advancing to the precise identification of enhancers and promoters along reference genome coordinates using the independent multi-omics assays. Our strategy addresses these challenges through a three-step integrative process. (1) Definition of candidate enhancer or promoter regions: Candidate regions are identified based on ChIP-seq peaks for H3K4me1, H3K4me3, and H3K27ac. (2) Identification of confident potential enhancer or promoter: Candidate peaks are intersected with additional CAGE-seq and ATAC-seq, retaining the peaks which overlap with at least one dataset and are present in at least two tissues. To avoid excluding strong individual signals, peaks with H3K4me1 or H3K4me3 signals with p values <0.0001 are also included. Both intersected and individually strong peaks are merged as confident potential enhancers or promoters. Promoter regions are further refined by integrating RNA-seq and CAGE-seq data to identify all potential transcription start sites (TSS), resulting in high-confidence promoters for each tissue. (3) Consistent genomic coordinates for confident potential enhancers or promoters: Potential enhancer and promoter regions are unified by consolidating overlapping regions across multiple tissues. The median start and end positions of overlapping peaks are calculated to define consistent genomic coordinates on the reference genome. This strategy effectively integrates independent datasets, ensuring precise and consistent annotation of enhancers and promoters across all tissues. By providing unified reference genomic coordinates, this approach represents a significant advancement in functional genome annotation, offering a robust framework for studying regulatory elements in the context of a reference genome.

To further demonstrate the robustness and reliability of our enhancer and promoter annotations, we generated and analyzed replicate ChIP-seq datasets for 16 representative tissues from the same individual sheep. These replicates, sequenced at higher depth, provided an independent evaluation of the reproducibility of our original data and the stability of our integrative analytical framework. Consistent with the original findings, the replicate datasets exhibited high sequencing quality and strong concordance across all evaluated metrics. Specifically, the mapping rates exceeded 98% for most tissues, and the proportion of duplicate reads was generally below 20%, indicating high library complexity and consistent experimental

performance across tissues (Supplementary Data 21). Even in tissues where duplicate rates were higher (e.g., up to ~50%), the number of uniquely mapped deduplicated reads remained above 20 million, which is widely accepted as sufficient for reliable peak calling and downstream regulatory analyses. Then, we assessed the reproducibility of regulatory element identification between the original and replicate datasets. Across the 16 tissues, enhancer regions derived from H3K27ac and H3K4me1 ChIP-seq showed a mean overlap rate of 62% with their corresponding replicates, while promoter regions defined by H3K27ac and H3K4me3 exhibited a higher concordance of approximately 82%, with some tissues exceeding 90% (Supplementary Table 5). These results clearly indicate that the regulatory elements identified in the original dataset are highly reproducible and biologically meaningful. Notably, the cerebellum and spinal cord tissues, which displayed particularly high reproducibility (enhancer overlap >80% and promoter overlap >90%), also demonstrated strong tissue-specific regulatory signatures consistent with neuronal gene expression patterns. To confirm that this reproducibility reflects biological consistency rather than technical redundancy, we performed comparative analyses using both Nextflow-based ChIP-seq pipelines and independent evaluations with Picard's MarkDuplicates module. Both analyses produced nearly identical mapping and duplication profiles, demonstrating that the reproducibility of our results is not pipeline-dependent. Moreover, the overlap between enhancers and promoters identified from independent omics datasets (CAGE-seq, ATAC-seq, and RNA-seq) further corroborated the reliability of our integrated enhancer-promoter framework. Candidate enhancers were retained only when supported by at least two independent datasets and detected in two or more tissues, minimizing the influence of variability from any single experiment. Together, these replicate-based validations provide compelling evidence that our enhancer and promoter annotations are both technically sound and biologically reliable. The consistent overlap between original and replicate datasets across multiple histone marks and tissues demonstrates that our multi-omics integration strategy effectively mitigates experimental variability and ensures reproducible detection of regulatory elements. This robustness strengthens the overall confidence in our dataset and supports its broad utility as a foundational resource for future studies on ruminant functional genomics, gene regulation, and evolutionary biology.

We combined experimental data on histone modifications (ChIP-seq), chromatin accessibility (ATAC-seq), and transcript expression and abundance (CAGE-seq and RNA-seq) to enable the de novo definition of transcriptional regulatory elements in the sheep genome. These regulatory elements, including promoters and enhancers, were annotated across 24 tissues from a single Rambouillet ewe, the source of DNA for the current reference genome assembly[20]. The use of multiple tissues from a single animal, rather than a few tissues from several individuals, was guided by four key considerations: (1) obtaining a broader understanding of tissue-specific differences was prioritized over achieving higher confidence in annotations within a limited number of tissues; (2) comparisons between tissues with similar expression profiles increased confidence in promoter, enhancer, and enhancer-promoter combination annotations; (3) using tissues from the reference animal ensures broad adaptability for researchers and facilitates timely updates, making the findings more universally applicable and valuable as a reference for other studies; and (4) biological replication within this breed is likely to be breed-specific, and replication efforts are more appropriately addressed by ongoing projects focused on specific breeds rather than as part of this annotation initiative.

Building upon the accurate identification of enhancers and promoters, cis-regulatory enhancer-promoter combinations were further defined by correlating enhancers located within a ±1 Mbp range of promoters with gene expression (TPM), retaining combinations with a correlation coefficient (r) ≥0.8. Through this approach, genes regulated by enhancers were identified, uncovering hidden regulatory mechanisms that influence gene expression. Functional enrichment analysis of tissue-specific genes associated with enhancers demonstrated superior results compared to using promoter-associated gene sets alone, providing a more accurate reflection of tissue-specific characteristics. This not only validated the reliability of our enhancer-promoter combination identification but also highlighted the regulatory role of enhancers in gene expression, which is stronger and more tissue-specific compared to promoters. Consequently, the number and tissue-specificity of enhancers, as well as their associated genes, exhibited greater variation across tissues compared to promoters. This discovery aligns with existing biological insights and previous findings, which demonstrate that enhancers exhibit higher tissue specificity than promoters[13,42]. While promoters often function across multiple tissues to initiate gene transcription, enhancers are more likely to be active in specific tissues. For example, in the case of the *BDNF* gene, expressed in both the cerebellum and cerebral cortex of the brain, its tissue specificity could not be identified based on promoters alone. However, the enhancer-promoter combination approach revealed that the enhancer *chr15: 57390520-57390685* is a cerebellum-specific enhancer, providing critical insights into its regulation. Additionally, a total of 554 enhancer-promoter combinations involving 68 genes, including *BDNF*, were identified as having tissue-specific enhancers. This study represents the genome-wide identification of tissue-specific enhancer-promoter combinations across multiple tissues in the sheep genome. It significantly enhances the annotation of the sheep reference genome and provides crucial regulatory elements for understanding gene regulation in sheep. These findings offer valuable contributions to the broader understanding of tissue-specific gene regulation and the functional genomics of this economically significant species.

Comparison of enhancers and promoters across sheep, cattle, pig, human, and mouse revealed both conserved and variable regulatory elements among these species. While gene expression signatures in specific tissue types tend to be conserved across mammals, regulatory elements exhibit greater species specificity. This aligns with previous reports that enhancer sequences change rapidly across mammalian evolution, despite maintaining similar gene expression profiles in comparable tissues[26]. Including sheep regulatory elements in this comparative analysis enhances our understanding of regulatory dynamics across mammalian species, contributing to a broader perspective on gene regulation. Additionally, a direct comparison between sheep and cattle regulatory annotations allowed us to identify potential ruminant-specific features shared by these two ruminant species. We identified 69 gene-associated enhancer-promoter combinations that were present in both sheep and cattle but absent in the non-ruminants (pig, mouse, and human), suggesting that these combinations may have a role in ruminant-specific biology. The significant KEGG pathways included thiamine metabolism (P value = 0.035), bacterial invasion of epithelial cells (P value = 0.0089), sphingolipid signaling pathway (P value = 0.022), gastric cancer (P value = 0.036), and tight junction (P value = 0.043) (Supplementary Data 22). These pathways involved genes such as *ELMO1*, *CTNNA3*, *AKT2*, *SGPL1*, *THTPA*, *RAPGEF6*, and *HSPA4*. Notably, these pathways are primarily associated with the microbial digestive environment of the rumen, a unique fermentation chamber found only in ruminants. The dynamic balance of rumen microorganisms, such as bacteria, is crucial for proper rumen function, underscoring the importance of these pathways in ruminant-specific digestive physiology.

This functional annotation provides a critical framework for evaluating genomic variations and interpreting their potential impact on biologically important complex traits in sheep. Variations in enhancer and promoter sequences have been demonstrated to influence phenotypes and regulate conserved processes, including mammalian development[26]. Although the number of QTLs reported in sheep is substantially lower compared to cattle, this study identified 38

QTLs colocalized in promoter regions of 17 genes and 215 QTLs overlapping with 214 enhancers. Notably, one QTL with a strong effect on milk fat yield, originally identified through a genome-wide association study (GWAS) of milk production traits[27], was pinpointed at chromosome 3 (*chr3*: *46457793*). This genomic position lies within a specific enhancer region (*chr3*: *46457598-46458034*) that interacts with *COMMD1*. *COMMD1* is expressed in at least 21 tissues and is a compelling candidate gene given its established role in copper metabolism, which has been linked to lipid metabolism and milk fat composition in ruminants[27,28]. These results suggest a plausible regulatory link between this enhancer, *COMMD1* expression, and milk fat yield, although additional functional validation (eQTL analyses or experimental perturbation) will be required to establish causality. More broadly, our findings demonstrate how the regulatory element annotations of the sheep reference genome provide a valuable context for understanding enhancer and promoter variations across individual sheep and breeds selected for diverse production traits. As more QTLs are discovered with the advancement of trait-related research, this regulatory element annotation will serve as a powerful tool for linking SNPs to critical traits, enabling a more precise interpretation of the biological significance of QTLs. This comprehensive dataset will facilitate deeper insights into the genetic mechanisms underlying economically important traits in sheep.

This study represents a significant step forward in the systematic identification and annotation of enhancers and promoters in the sheep genome, providing a unified framework for understanding the transcriptional regulatory landscape across multiple tissues. By transitioning from traditional cell/tissue-peak approaches to the incorporation of a multi-omics faceted strategy, we integrated multi-omics data to achieve precise mapping of enhancers and promoters to unified genomic coordinates. This strategy not only enhances the functional annotation of the sheep reference genome but also provides a foundation for exploring the regulatory mechanisms underlying complex traits and tissue-specific gene expression. While rigorous, this approach may have been overly stringent, potentially excluding some true enhancers or promoters. As such, the current findings could be further refined and expanded with additional data and advanced experimental and computational approaches. Future improvements might include integrating more specific and high-resolution experimental data, such as Hi-C across multiple tissues, for enhancer and promoter identification, as well as leveraging artificial intelligence models trained on diverse and comprehensive datasets. These advancements hold the potential to enhance the accuracy of genome-wide predictions of regulatory elements, ultimately providing a more complete and nuanced understanding of gene regulation in sheep and other species.

Methods

Tissue collection

Tissues were collected from a healthy 6-year-old Rambouillet ewe (Benz 2616, USMARC ID 200935900), whose genome was sequenced and assembled de novo to produce the ovine reference genome ARS_UI_Ramb_v2.0[20]. Ethics approval for all animal procedures were obtained by the Utah State University Institutional Animal Care and Use Committee on April 29, 2016, at the Utah Veterinary Diagnostic Laboratory. Postmortem tissue samples were immediately transferred to cryogenic vials and snap frozen in liquid nitrogen, with subsequent transfer to −80 °C freezers for storage. Further details regarding the tissue collection strategy are available through the FAANG Data Coordination Center's data portal (https://data.faang.org/home), and the metadata are recorded according to FAANG guidelines under BioSample accession SAMEG329607[15,43].

Chromatin immunoprecipitation with sequencing (ChIP-seq)

Chromatin immunoprecipitation (ChIP) was performed for 24 collected tissues from Benz 2616 (Supplementary Table 1). The SimpleChIP Plus Enzymatic Chromatin IP Kit (Cell Signaling Technologies, cat. #9005, Danvers, MA, USA) was used for immunoprecipitation according to the manufacturer's instructions[44,45]. Antibodies specific for histone modifications H3K4me3 (Abcam, cat. #ab8580), H3K27ac (Abcam, cat. #ab4729), H3K4me1 (Abcam, cat. #ab8895), and H3K27me3 (Abcam, cat. #ab6002) were used for immunoprecipitation. Briefly, tissue was crosslinked with 37% formaldehyde, disaggregated with Dounce homogenization, and chromatin was sheared with micrococcal nuclease (MNase) by incubating on a thermomixer for 20 min at 200 rpm and 37 °C. Following nuclear membrane lysis and isolation of sheared chromatin by centrifugation at 15,000×*g* for 1 min, chromatin was incubated with 1 μg of antibody overnight (16 h) at 4 °C on a HulaMixer (Thermo Fisher Scientific, USA). An input control (20 μl from each immunoprecipitation) sample was removed from sheared chromatin prior to incubation with antibodies and stored at −20 °C until purification. Protein-G-coated magnetic beads were used to isolate chromatin bound to specific antibodies, and purification of enriched chromatin was performed with the DNA Purification Buffers and Spin Columns Kit following manufacturer's instructions (Cell Signaling Technologies, cat. #14209, Danvers, MA, USA).

Purified DNA was quantified using a Qubit dsDNA HS Assay Kit (Thermo Fisher Scientific, cat. #Q32854, Waltham, MA, USA), and size and integrity was verified with a Fragment Analyzer (Agilent, Santa Clara, CA, USA). Libraries were prepared as 75 bp paired end reads with the TruSeq ChIP Library Preparation Kit (Illumina, Inc., cat. #IP-202-1012, San Diego, CA, USA) following manufacturer's instructions. Libraries for H3K4me3, H3K27ac, and Input were sequenced to achieve at least 20 million uniquely mapped reads, and H3K4me1 and H3K27me3 libraries were sequenced to achieve at least 45 million uniquely mapped reads to comply with ENCODE standards for ChIP-seq[1].

Quality control of sequence data was performed with FastQC v0.11.9 and MultiQC v1.14[46], and the peaks of H3K4me3, H3K27ac, H3K4me1, and H3K27me3 were called by the chipseq model of Nextflow v24.04.2[47] with Bowtie2 aligner v2.4.2 and MACS2 peak caller v2.2.7.1 with "-macs_pavalue 0.005"[48,49], for narrow histone marks H3K4me3 and H3K27ac with additional parameter "--narrow_peak". The reference genome for alignment was from NCBI (ARS-UI_Ramb_v2.0, GenBank accession GCF_016772045.1)[20], which was also used for all other assays, and the annotation was generated by the standard NCBI annotation pipeline v9.0. The genomic distribution of H3K4me1, H3K27me3, H3K4me3, and H3K27ac peaks across feature regions, including promoters (TSS ± 2 kb), exons, introns, and intergenic regions, was analyzed using the R packages GenomicFeatures v1.50.4[50] and ChIPseeker v1.34.1[51].

Cap analysis gene expression sequencing (CAGE-seq)

CAGE-seq data from 24 tissues was generated from ref. 15, and the libraries were prepared for each sample from a starting quantity of 5 μg of DNase-treated total RNA. Random primers were used to ensure conversion of all 5′-cap-trapping RNAs according to ref. 15. The full protocol is available via the FAANG Data Coordination Centre (https://data.faang.org/protocol/experiments/cageProtocol-CAGE-seq-TSS). CAGE-seq data were used to identify the transcription start site (TSS) for each gene. The analysis pipeline and metrics are available on Bitbucket (https://bitbucket.org/msalavat/cagewrap_public/src/master). Briefly, raw CAGE reads were trimmed using tagDust2[52] and mapped to the ARS-UI Ramb v2.0 using bowtie2 v2.4.2 (--very-sensitive mode)[49]. The mapped BAM files were converted to strand-specific bedGraphs using BEDTools v2.29.0[53] and subsequently converted to bigWig format using the UCSC BedGraphToBigWig tool[54]. The clustering analysis was carried out in R using CAGEfightR v1.12.0[55], applying "unidirectional" and "bidirectional" clustering algorithms to identify putative TSSs and TSS-Enhancer regions, respectively. The interaction between

TSSs and enhancers were further determined by the findLinks function in CAGEfightR v.1.12.0[55].

Assay for transposase-accessible chromatin with sequencing (ATAC-seq)

The assay for transposase accessible chromatin (ATAC) was performed on 20 tissues as a service provided by the University of California, San Diego Center for Epigenomics. Libraries were sequenced to a depth exceeding 90 million reads (Supplementary Data 1). Quality control and peaks calling parameters for ATAC-seq were consistent with those used for ChIP-seq, utilizing the atacseq model of Nextflow V24.04.2[47].

RNA sequencing

RNA was isolated from snap-frozen tissue by cryopulverizing tissue and placing it in 2 mL of Trizol. The tissue was then homogenized in the Trizol and split into 1 mL aliquots in 1.5 mL microcentrifuge. Samples were then centrifuged for 5 min at 14,000×*g* and 4 °C. The aqueous (clear) layer was transferred to a new tube, avoiding any fat that may have risen to the top of the liquid. Then, 400 μL of chloroform was added to each tube, shaken vigorously, and incubated at room temperature for 15 min. The sample was centrifuged for 15 min and 14,000×*g* and 4 °C, followed by transfer of the aqueous layer into another 1.5 mL microcentrifuge tube. About 500 μL of isopropanol was then added to each tube and incubated for 10 min at room temperature, followed by centrifugation at 14,000×*g* and 4 °C for 30 min. The supernatant was discarded, and the pellet was washed with 75% ethanol. The purified RNA was resuspended in nuclease-free water and quantified with a Nanodrop, as well as quality evaluated with a Fragment Analyzer. RNA was then treated with DNase I (1 μL of 2000 U/mL per 10 μg RNA) and cleaned up with the Rneasy MinElute spin column kit (Qiagen).

RNA-seq libraries were prepared using the Illumina TruSeq Stranded mRNA Library Preparation Kit (Illumina, Inc.) according to the manufacturer's instructions. Sequencing was performed on an Illumina NextSeq 500, generating at least 65 million reads per sample (Supplementary Data 1). This sequencing depth ensures that fragments of most expressed genes are covered at least 30-fold, providing sufficient confidence for expression quantification. The reads were aligned to the ARS-UI Ramb v2.0 reference genome using STAR v2.7.8a[56]. Gene expression levels were quantified as transcripts per million (TPM) using StringTie v2.2.0[57], and genes with TPM ≥0.5 were classified as expressed only for the quantitative analysis of enhancer-promoter expression correlations, while no expression filtering was applied in subsequent qualitative comparisons.

Chromatin state characterization

Chromatin states for the 24 tissues were characterized using the identified ChIP-seq and ATAC-seq peaks and analyzed with the Hidden Markov Model in ChromHMM[21], which integrates overlapping histone modification signals and chromatin accessibility to define chromatin states, including active, poised, repressed promoter and enhancer and repressor. The optimal number of chromatin states was examined by implementing ChromHMM with 2 through 16 chromatin states in the *LearnModel* function, followed by the median Pearson correlation output by the *CompareModels* function, where each tissue plateaued and was tightly correlated with the model with the greatest number of states[37,58]. The percentage of the genome occupied by each chromatin state was calculated and averaged across tissues, providing a comprehensive view of chromatin landscape variability.

Identification of enhancers and promoters

Histone modifications have been widely used to characterize enhancers and promoters[10,41]. Specifically, H3K4me3 and H3K27ac are commonly associated with promoters, while H3K4me1 and H3K27ac indicate enhancers. To achieve a more reliable and accurate identification of enhancers and promoters, we developed a comprehensive strategy integrating four distinct high-throughput omics datasets: ChIP-seq, CAGE-seq, ATAC-seq, and RNA-seq. The 5′ end of the TSS and enhancer RNA were characterized by unidirectional and bidirectional sequences (CAGE-unid and CAGE-id) from the CAGE-seq data, respectively. Chromatin accessibility was mapped using ATAC-seq data. The TSS and active gene expression were profiled by RNA-seq (Fig. 2A).

The specific processes of this strategy are:

I. Definition of candidate enhancer or promoter regions for each tissue:

$$Peak_{\mathrm{enh}} = P_{\mathrm{H3K4me1}} \cup P_{\mathrm{H3K27ac}} \tag{1}$$

$$Peak_{\mathrm{prom}} = P_{\mathrm{H3K4me3}} \cup P_{\mathrm{H3K27ac}} \tag{2}$$

Where $Peak_{\mathrm{enh}}$ and $Peak_{\mathrm{prom}}$ are candidate peak regions for enhancer and promoter, respectively, and $P.$ represents peaks from the respective histone marks from ChIP-seq data.

II. Identification of confident potential enhancers or promoters:

To refine candidate regions, additional datasets were integrated. For each tissue (T_i), the potential enhancers and promoters were defined as follows:

$$P_{\mathrm{enh_Ti}} = (Peak_{\mathrm{enh_Ti}} \cap P_{\mathrm{CAGE-bid_Ti}}) \cup (P_{\mathrm{H3K4me1_Ti}} \cap P_{\mathrm{H3K27ac_Ti}}) \cup (P_{\mathrm{H3K4me1_Ti}} \cap P_{\mathrm{ATAC_Ti}}) \tag{3}$$

$$P_{\mathrm{prom_Ti}} = (Peak_{\mathrm{prom_Ti}} \cap P_{\mathrm{CAGE-unid_Ti}}) \cup (P_{\mathrm{H3K4me3_Ti}} \cap P_{\mathrm{H3K27ac_Ti}}) \cup (P_{\mathrm{H3K4me3_Ti}} \cap P_{\mathrm{ATAC_Ti}}) \tag{4}$$

Given ATAC-seq data availability for only 20 tissues, $i = 1, 2, \ldots, 20$. The union of confident peaks across all tissues yielded:

$$P_{\mathrm{enh_all}} = \bigcup_{i=1}^{20} P_{\mathrm{enh_Ti}}, P_{\mathrm{prom_all}} = \bigcup_{i=1}^{20} P_{\mathrm{prom_Ti}} \tag{5}$$

We used $P_{\mathrm{enh_all}}$ and $P_{\mathrm{prom_all}}$ as the potential enhancer and promoter regions for filtering across all 24 tissues. For a given i-th tissue (T_i), the any potential enhancer (Enh_{ij}) and promoter ($Prom_{ij}$) regions can be represented as follows:

$$\mathrm{Enh}_{ij} \in P_{\mathrm{enh_all}}, \mathrm{Prom}_{ij} \in P_{\mathrm{prom_all}} \tag{6}$$

Where j is the j-th enhancer or promoter from tissue i (T_i), respectively. To refine potential enhancer and promoter regions, we applied two criteria:

1 The region must be identified in at least two tissues ($\sum_{i=1}^{24} t\left(Enh_{ij}\right) \geq 2$), where $t(.)$ is an indicator function:

$$t(x) = \begin{cases} 1 & \text{if } x \in P_{\mathrm{enh_all}} \text{ or } P_{\mathrm{prom_all}} \\ 0 & \text{otherwise} \end{cases} \tag{7}$$

2 The region must have a p value ($Pval_{\mathrm{H3K4me1}}$ for enhancers, $Pval_{\mathrm{H3K4me3}}$ for promoters) below 0.001.

The final confident enhancer and promoter regions for each tissue were defined as:

$$\mathrm{Enh}_{Ti} = \left\{ \mathrm{Enh}_{ij} \in P_{\mathrm{enh_all}} \middle| \sum_{i=1}^{24} t\left(\mathrm{Enh}_{ij}\right) \geq 2 \text{ or } Pval_{\mathrm{H3K4me1}}\left(\mathrm{Enh}_{ij}\right) \leq 0.001 \right\} \tag{8}$$

$$\mathrm{Prom}_{Ti} = \left\{ \mathrm{Prom}_{ij} \in P_{\mathrm{prom_all}} \middle| \sum_{i=1}^{24} t\left(\mathrm{Prom}_{ij}\right) \geq 2 \right. \\ \left. or\ Pval_{\mathrm{H3K4me3}}\left(\mathrm{Prom}_{ij}\right) \leq 0.001 \right\} \quad (9)$$

Taken together, the potential enhancer and promoter regions were defined using integrative evidence from histone modification profiles (H3K4me1, H3K27ac, and H3K4me3), CAGE bidirectional and unidirectional transcription signals (CAGE_bid and CAGE_unid), and ATAC-seq chromatin accessibility data (ATAC) (Supplementary Table 6). Promoters were further refined using transcription start site (TSS) information (±2 kb of TSS). TSSs were derived from two sources: (1) annotated genes in the reference genome, and (2) unannotated genes from RNA-seq and CAGE-seq. identified unannotated TSSs were extracted from StringTie-assembled transcripts with TPM ≥10, overlapping with CAGE-seq TSS regions, and detected in at least two tissues.

III. Consistent genomic coordinates for enhancers and promoters:

To unify the genomic coordinates of enhancers and promoters across tissues, overlapping regions in Enh_{Ti} or Prom_{Ti} from multiple tissues were consolidated using a stepwise median approach. For k-overlapping tissues ($k \leq 24$), with start positions $\{s_1, s_2, \ldots, s_k\}$ and end positions $\{e_1, e_2, \ldots, e_k\}$, the consolidated region $[S_m, E_m]$ was calculated iteratively:

$$S_m = \begin{cases} s_{(k+1)/2} & \text{if } k \text{ is odd} \\ (s_{k/2} + s_{(k/2+1)})/2 & \text{if } k \text{ is even} \end{cases}, \quad E_m = \begin{cases} e_{(k+1)/2} & \text{if } k \text{ is odd} \\ (e_{k/2} + e_{(k/2+1)})/2 & \text{if } k \text{ is even} \end{cases} \quad (10)$$

Regions outside the interval $[S_m, E_m]$ were iteratively removed, and the process was repeated until no points remained outside. The resulting unified enhancer or promoter regions were $\{[S_{m,i}, E_{m,i}]\}_{i=1}^{k}$, where k is the number of unique enhancer or promoter regions.

This approach ensured a consistent reference genome annotation for enhancers and promoters, integrating signals from multiple tissues into a unified framework.

Cis-regulatory elements of enhancers and promoters on genes

To investigate cis-regulatory interactions between enhancers and promoters, all enhancers within ±1 Mbp of transcription start sites (TSS) were identified based on their genomic coordinates. Potential enhancer-promoter combinations were predicted using three complementary approaches. First, enhancer-gene combinations were identified by calculating the Spearman's correlation ($r \geq 0.8$) between the tissue-specific activity profile of each enhancer (presence or absence across 24 tissues) and the corresponding gene expression (TPM) across the same tissue. Second, CTCF ChIP-seq data from cerebellum and liver tissues, collected in a previous study[7], were used to identify CTCF-mediated chromatin loops. CTCF peaks were overlapped with CTCF motifs identified using FIMO to determine CTCF orientation[59,60]. Using this data, loops were identified following the pipeline from ref. 61. Loop containing single promoter and enhancers from the same cluster were used as candidate enhancer-promoter cis-regulatory combinations and validated across all 24 tissues. Third, enhancer-promoter combinations were identified based on promoter and enhancer links from CAGE data using the CAGEfightR v1.12.0[55]. These potential combinations from the three sources were combined, and those with an FDR < 0.05 (Benjamini-Hochberg correction) were retained as confident enhancer-promoter interactions. For each tissue, TPM values were compared between expressed genes with identified enhancers and those without, to evaluate the regulatory impact of enhancers. Additionally, enhancer-promoter interactions were categorized into two primary patterns: enhancer hubs (based on the number of promoters associated with a single enhancer) and promoter hubs (based on the number of enhancers associated with a single promoter). This comprehensive integration of multiple datasets provides a robust framework for elucidating tissue-specific cis-regulatory mechanisms.

Tissue-specific enhancer and promoter

To investigate tissue-specific regulatory elements, we identified and compared genes linked to tissue-specific enhancers and promoters derived from enhancer-promoter combinations. Gene Ontology (GO) enrichment analysis was conducted on genes associated with tissue-specific enhancers and compared to those linked to tissue-specific promoters using Enrichr-KG (https://maayanlab.cloud/enrichr-kg)[62]. Additionally, tissue-specific enhancer sequences were analyzed to identify associated transcription factors (TFs) using the HOMER annotation tool (http://homer.ucsd.edu/homer) with the annotatePeaks.pl function[63]. This approach provides insight into the functional roles and regulatory networks associated with tissue-specific enhancers and promoters.

Comparisons of enhancer and promoter across five species

To analyze species-specific regulatory elements in sheep, we compared ChIP-seq data with four other mammalian species, including cattle, pig, mouse, and human. Histone modification data from four tissues (cerebellum, cerebral cortex, lung, and muscle) were used for this comparison. Data for cattle[64], pig[9], mouse (ENCSR738IWZ), and human (ENCSR000FYQ) were sourced from publicly available datasets. Promoter regions in these species were defined as regions overlapping a flanked TSS region (upstream 2.5 kb and downstream 1 kb)[65]. And the annotated enhancer regions of cattle, pig and mouse were downloaded from (GIGA)^{n}DB[66], while human enhancer regions were obtained from the EnhancerAtlas 2.0 database[67].

Whole-genome alignment chain files for human (hg38), mouse (mm10), pig (SusScr11), and cattle (bosTau9) were downloaded from the University of California, Santa Cruz (UCSC) database. The chain files processed using reciprocal best chain methods from the UCSC Genome Wiki (http://genomewiki.ucsc.edu/index.php). Genomic coordinates were converted between the sheep reference and other mammal assemblies using the UCSC Liftover tool (https://genome.sph.umich.edu/wiki/LiftOver) with a threshold of 0.8 sequence identity. Human enhancer and promoter regions were used as a reference, and annotations for sheep, cattle, pig, and mouse were lifted over to human coordinates for direct comparison. Shared enhancers and promoters from both sheep and cattle were identified as the ruminant-specific regulatory elements. The KEGG pathway analysis of genes associated with ruminant-specific enhancers and promoters were performed using KOBAS 3.0[68]. Pathways with p value <0.05 were considered significant.

Biological insights into enhancers and promoters

Whole-genome sequencing (WGS)-based SNPs of 355 sheep were obtained from the iSheep database (https://ngdc.cncb.ac.cn/isheep/download), which contains 15,816,440 SNPs mapped to ARS_UI_Ramb_v2.0. The genomic distribution of these SNPs was analyzed across exonic, intronic, promoter, and enhancer regions to identify their functional implications. To explore the biological functions of SNPs within enhancer and promoter regions, we retrieved sheep QTL data from the sheepQTLdb (Release 57, Aug 26, 2025; https://www.animalgenome.org/QTLdb), which includes 8,288 QTL records curated from 293 publications covering 182 base traits and 266 trait variants, providing valuable insights into the functional significance of regulatory elements in sheep. Additionally, birth weight-associated SNPs (28,736 variants with P value ≤0.001) identified from a genome-association study (GWAS) of 3007 sheep were analyzed, together with 181 eQTL loci, to further validate the reliability and biological relevance of the identified regulatory elements[29].

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data supporting the findings of this work are available within the article, Supplementary Information files, Source Data file and Faangsheep [https://bioxsh.wixsite.com/faangsheepncsub]. The read statistics from all datasets across 24 tissues in this study are provided in Supplementary Data 1. The enhancers and promoters identified across all tissues are provided in Supplementary Data 3 and Data 4, respectively. The peaks from ChIP-seq data are available and Supplementary Data 2. The ChIP-seq (H3K4me3, H3K27ac, H3K4me1, and H3K27me3) and CTCF data were available under BioProject PRJEB41457. The CAGE-seq data generated in this study have been deposited in the ENA database under project number PRJEB34864. The RNA-seq data used in this study have been deposited in the ENA database under project number PRJEB35292. The methylation data have been deposited in the ENA database under project number PRJEB39178. The ATAC-seq generated in this study have been deposited in the SRA database under PRJNA1266684. Previously published GWAS and eQTL data were available in Zenodo [https://doi.org/10.5281/zenodo.15696310]. Source Data are provided with this article. Source data are provided with this paper.

Code availability

All the bioinformatic scripts used in this study are available at the CodeOcean repository with https://doi.org/10.24433/CO.4241215.v1 and GitHub: https://github.com/shang-qian/FAANG_sheep_code_NCsub under the MIT License.

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Author contributions

N.E.C, S.N.W., E.L.C., A.L.A., S.M.C., K.C.W., T.P.L.S., and B.M.M. designed and supervised the study. S.X., S.D.M., and B.M.M. conceived and developed the methodology. S.X., K.M.D, B.D.R., B.M.M., M.S., E.L.C., A.C., R.B., S.M.C, and K.C.W. performed the data analyses. K.M.D, B.M.M., T.P.L.S., K.C.W., T.H., M.S., E.L.C., A.T.M., M.R.M, S.N.W., A.C., S.M.C., M.R.S., S.F.K, K.A.S., and T.A.O. performed the laboratory work. S.X., K.M.D., A.C., S.M.C., M.S., G.B., and B.M.M. wrote the manuscript with main contributions from T.P.L.S, E.L.C., A.L.A, G.K.M., and S.D.M. The manuscript was revised by S.X., K.M.D., S.D.M., and B.M.M. All authors reviewed and approved the final version of the manuscript.

Competing interests

The authors declare no competing interests

Additional information

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Correspondence and requests for materials should be addressed to Brenda M. Murdoch.

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